A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 27

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines – no generation when levels are equal (slack water) **OR** possible on the flood when the sea is higher than the basin **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow – the timing does not match demand **1** Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1 Spring tides:** Sun and Moon aligned so their gravitational effects combine giving a large tidal range – neap: effects oppose giving a small range **1** Energy depends on the mass of water and height difference, so output is much greater at springs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** When demand is low, surplus electricity pumps water into the lagoon above sea level **1** The stored water is released through the turbines later when demand is high (pumped storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak	2
01.5	**1 mark each** 1 mark for each of the 2 points below. **1** Large tidal range (eg Severn Estuary) **1** Estuaries, bays or headlands / channels that focus the flow and increase velocity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1.	
02.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Stress-related illness:** ulcers, high blood pressure, heart disease **1 Behavioural changes:** irritability, aggression, lack of concentration leading to accidents **1** Communication problems / educational underperformance (eg schools under flight paths) **1** Sleep disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1 Acoustic fatigue:** objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) **1 Vibration damage:** repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes **1 Shock impacts:** the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Birds that sing to establish territories / attract mates cannot be heard against background noise **1** Birds scared by sudden noise leave nests so predators take eggs / chicks – cliff-nesting seabirds knock eggs or chicks into the sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Whales / dolphins use hearing / echolocation to find food and communicate – loud sonar causes hearing damage if close to the source **1** Behavioural changes / disorientation possibly linked to mass strandings – reduced feeding success **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Deforestation:** less interception so less evaporation from leaves – less transpiration so less water returned to the atmosphere and less precipitation downwind – more infiltration and surface runoff, higher river flows / flooding **2 Urban development:** impermeable concrete / tarmac reduce infiltration and aquifer recharge – runoff is faster and greater, increasing river flooding downstream – drains carry water quickly to rivers **2 Agriculture:** soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff – irrigation increases evaporation – loss of soil biota (worms) reduces infiltration and water retention – drainage lowers water tables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **infiltration** → water soaking into the soil surface • **impermeable** → does not let water through **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	6
03.2	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation **1** More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods **1** Changed wind patterns move humid air to different areas **1** Sea level rise / salt incursion into coastal aquifers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity **1** Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well **1** Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well **1** Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **hydraulic fracturing** → pumping fluid into rock at high pressure to release gas or oil • **permeability** → how easily water passes through / lets water through	3
04.2	**1 mark each** 1 mark for each of the 5 points below. **1** Natural gas may enter aquifer water **1** Injected chemicals may reach aquifers or the surface – toxic metals in the rock may be mobilised **1** Large volumes of water needed **1** Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally) **1 Reduced by:** collection, treatment and re-use of waste water – well casing standards – restricting sites away from aquifers / sensitive areas – monitoring **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **fracking** → pumping fluid into rock at high pressure to release gas or oil • **aquifer** → a permeable rock that stores groundwater / underground water store	5
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices **1** Reserves only include what is economically exploitable now – high environmental impacts (land take, water, CO_2) also limit exploitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Inorganic mercury sank into the sediments / deep water where conditions are anaerobic / low oxygen **1** It was chemically reduced / converted by (anaerobic) bacteria to methyl mercury **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Methyl mercury is liposoluble so it passes through cell membranes and is stored in fat / not excreted **1** So it bioaccumulates in organisms and biomagnifies along the food chain (eg in fish, then humans eating fish) **1** It is a neurotoxin / teratogen causing brain damage and birth abnormalities (eg Minamata) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Mercury is persistent / not degraded – it is an element so cannot be broken down **1** It remains in the sediments and continues to be released / converted and taken up by the food chain – long-lived fish continue to accumulate it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Water current velocity and direction (dispersal) **1** **Properties of the pollutant:** solubility, persistence, liposolubility / potential to bioaccumulate **1** Food chain pathways to humans (fish, shellfish) / the critical group **1** Sediment type / adsorption onto clay **1** Volume of the water body / residence time / dilution **1** Sensitive locations downstream (fisheries, drinking water abstraction) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **critical group** → the people most at risk / most heavily exposed • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 238-242; AQA spec 3.4.1, 3.4.2.	
06.1	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Discharge of tank-washing water from tankers **1** Oil rig / drilling accidents / leakage during drilling **1** Pipeline leaks **1** Leakage from storage tanks / refinery spills **1** Waste lubricating oil from vehicles / industry washed into drains **1** Other ship accidents (fuel oil) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.2	**1 mark each** 1 mark for each of the 3 points below. **1** Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia **1** Feathers lose buoyancy / waterproofing so heavily oiled birds drown **1** Birds swallow toxic oil when preening – time spent cleaning feathers reduces time for feeding themselves / their young **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen **1** Toxic components (benzene, xylene) dissolve and poison plankton near the surface **1** Oil blocks light so algae / phytoplankton cannot photosynthesise **1** Oil masks odours / narcotises so animals cannot find mates or food by smell **1** Shore organisms (molluscs, crustaceans) are covered and asphyxiated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Dispersal / dilution by currents and waves is prevented so oil stays concentrated – in open water most spills disperse in days or weeks **1** Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Low liposolubility and low persistence **Also accept** • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1 Bioaccumulation:** build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown **1 Biomagnification:** increase in concentration of the pollutant at successive trophic levels of a food chain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **bioaccumulation** → a pollutant building up inside one organism over time • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are intended to be toxic (to the target species) **1** They are deliberately released / spread into the environment (rather than being an accidental or waste release) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
09.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (2.9 - 0.7) / 2.9 x 100 = 75.9% (accept 76%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Although the larger vehicle uses more energy in total, it carries many more people / a larger load **1** So the energy per passenger (or per tonne of goods) is lower – steel wheels on rails also have low rolling resistance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Flexibility of routes / destinations and door-to-door convenience **1** Time / speed and reliability where public transport is infrequent **1** Low fuel tax / cheap fuel **1** Carrying goods – comfort – status **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 4 points below. **1** Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey **1** So more journeys shift from cars to bulk transport / cycling **1** ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely **1** Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** t-test (or Mann-Whitney U if not normally distributed) **1** Comparing two means of counted / measured data from two independent samples **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **t-test** → a test for a difference between the means of two normally distributed sets • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** The difference is statistically significant (less than 1% probability that it arose by chance) **1** The null hypothesis (no difference between fields) is rejected – the organic field has significantly more earthworms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** A significant difference does not show a causal relationship **1** Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **causal** → one variable actually causing the change in the other **Do not credit** • 'organic farming uses no pesticides' - a restricted list of pesticides is permitted	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Scientific methodology:** identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research • **Sample location:** random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias – interval chosen so variation is detected without excess data • **Sample timing:** diurnal, seasonal, weather-related and long-term changes – sampling on different occasions – interval matched to rate of change • **Sample size and number:** larger and more samples for non-homogeneous variables – running mean and standard deviation show when the mean is reliable – preliminary study to decide • Standardisation and control of variables – accuracy, precision, reliability, representativeness – anomalies and repeats – calibration of equipment – appropriate techniques and risk management • **Statistical analysis:** choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation • **Evaluation:** trade-offs between cost, time, disturbance and reliability – limitations of sub-sampling – justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **valid** → actually measures what it is meant to measure • **null hypothesis** → a statement that there is no significant difference or correlation • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	
	Model answer - Level 3 (8 marks)	
	A valid investigation begins with a clear, testable hypothesis and a null hypothesis, informed by existing knowledge, so that the data collected can actually accept or reject it. The first design decision is sample location: random sampling (grid coordinates or numbered sites) or systematic sampling along transects avoids the bias that would come from choosing convenient sites, and a preliminary study shows the interval needed to detect variation without collecting unnecessary data.	
	Timing matters because many variables change diurnally, seasonally or with weather, so samples on different occasions are needed and the interval must match the rate of change. Sample size and number depend on how variable the population is: a running mean shows when enough samples have been taken and the standard deviation shows the scatter around the mean. Techniques must be standardised so results from different sites, times or researchers are comparable, other variables controlled or monitored, and equipment calibrated so that precise readings are also accurate; anomalies should be repeated rather than ignored.	
	Finally the analysis must be chosen to suit the data - Spearman's rank for correlation, chi-squared for frequencies, a t-test for two means or Mann-Whitney U for medians - so that significance ($p < 0.05$) can be assessed, remembering that a significant correlation does not prove causation. Every decision trades reliability against cost, time and disturbance; sub-samples can only estimate the whole, so the most valid study is the one whose sampling design makes its limitations explicit.	
	Weaker answer - Level 2 (5 marks)	
	When planning fieldwork you need a hypothesis. You should choose sites randomly so it is not biased, for example using random numbers. You need to take enough samples and repeat them so the results are reliable. The method should be the same each time. You should measure accurately with calibrated equipment. Then you do a statistical test such as a t-test to see if the results are significant. If p is less than 0.05 the result is significant. This makes the conclusions valid.	
	What makes the difference	
	• Level 2 is sound but generic: it says 'enough samples' where Level 3 explains how a running mean and standard deviation tell you how many.	
	• Level 3 covers timing and the choice between statistical tests; Level 2 mentions one test only.	
	• The evaluation is missing from Level 2 - there is no discussion of trade-offs or of correlation vs causation, so it cannot reach the top level however accurate it is.	

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	
11.1	**Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Hydrological cycle processes and reservoirs – residence times – dynamic equilibrium – increasing demand (population, affluence, irrigation, industry) • **Human impacts on transfer rates:** deforestation, agriculture, urbanisation, climate change, abstraction – consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • **Case studies of unsustainable exploitation:** Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • **Sustainable strategies:** metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair – new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis – pollution control (sewage, AMD, buffer strips)	25

Do not write outside the box

Question	Marking guidance	Marks
	• Links to thermohaline circulation and climate – energy costs – circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	
11.2	**Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility – DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • **Reducing impact:** restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • **Alternatives:** cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • **Trade-offs:** yield, cost, human health (organophosphates), resistance evolution, pollinator services – risk:benefit – sustainability and the circular economy	25

Do not write outside the box

Question	Marking guidance	Marks
	• **Judgement:** no pesticide is perfectly specific – combination of property-based selection and integrated control is most sustainable *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	

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